- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

**13** A projectile is fired from ground level at $35\,\text{m s}^{-1}$ at an angle of $\theta°$ above the horizontal.

**(a)** State a modelling assumption that is used in the standard projectile model. **[1]**

**(b)** Find the cartesian equation of the trajectory of the projectile. **[4]**

The projectile travels above horizontal ground towards a wall that is 110 m away from the point of projection and 5 m high. The projectile reaches a maximum height of 22.5 m.

**(c)** Determine whether the projectile hits the wall. **[6]**

**13(a)**

**13(b)**

**13(c)**